

Appendix A

Stage A Reproducibility Appendix

A.1 Scope

This appendix records the committed Stage A evidence needed to reproduce the paper-matched Molnar candidate-v2 benchmark review without launching a new Abaqus or PBS job. It is a provenance package, not a Gate A3 pass.

A.2 Repository Revisions

Table A.1: Relevant committed revisions.

Purpose	Revision
Current supervisor package	a7294b30851d7190ee91c8bf92390bf32b2442c8
Candidate-v2 generation, static validation, and baseline submission	711dd495bdc830d695f9d7e56283316c9d417d5
SDV15 diagnostic r2 preparation and submission	209ad325d2c85532411c13d8290db08ca35b0637

A.3 Job Records

Table A.2: Stage A HPC jobs used in the paper-matched candidate-v2 evidence set.

Job	Submitted revision	Result
1374864.mmaster02	711dd495bdc830d695f9d7e56283316c9d417d5	Paper-matched candidate-v2 baseline. PBS Exit.status=0; Abaqus return code zero; classification paper_matched_v2.technical_pass.
1375020.mmaster02	efd5f60ebb9cc6ea8ce89b508a6e9ff418345615	SDV15 diagnostic attempt. Failed before Abaqus due to missing git in batch PATH; no solver result.
1375028.mmaster02	209ad325d2c85532411c13d8290db08ca35b0637	Prepared SDV15 diagnostic r2. Abaqus technical pass; wrapper postprocessing failed under cluster Python compatibility; completed-increment replay and severity audit were performed without a new solution run.

A.4 Software and Runtime Environment

The candidate-v2 baseline used modules `gcc/11.4.0`, `intel/2024.2.0`, and `abaqus/2023` on the TU Freiberg HPC system. It ran serially with one CPU. The diagnostic r2 replay provenance records Abaqus Python as 2.7.15 with GCC 8.2.1.

Table A.3: Resource records.

Quantity	Baseline job	Diagnostic r2 job
PBS job ID	1374864.mmaste02	1375028.mmaste02
Queue	entry_imfdmkmq request	normal_imfdmkmq initial state
CPU request	1 CPU	1 CPU
Memory request	32gb	32gb
Walltime request	24:00:00	16:00:00
Used walltime	00:38:38	00:42:27
Used CPU time	00:35:52	not retained in run summary
Used memory	2970760kb; vmem 3565868kb; Abaqus peak about 2 GB	not retained in run summary
Execution host	mnode099/0	mnode100/0

A.5 Deck and Source Hashes

Table A.4: Candidate-v2 source and deck hashes.

Artifact	SHA-256
models/generated/molnar_ gravouil_2017/paper_matched_ single_notch_v2/paper_ matched_single_notch_v2.inp	f4d135d6c12d42a94c1874a6453a8865b4806a4e2d3b5018141471ba9245ecf2
models/baseline_original/ molnar_gravouil_2017/ 02_Single_Notch_Tension/ SingleNotch.for	18944e5bb2a3b7973fd0d4bff03f8e078eef667965343d8a29156d093f53f5f1
models/generated/molnar_ gravouil_2017/paper_matched_ single_notch_v2/SingleNotch_ v2.for	e587c195c50a5c52a000ca54d3ed00e1bf8161ba2f6aa7b3cf5aacec8425913a

A.6 Static Validation and Diagnostic Targets

Candidate v2 static validation is recorded as `static_validation_pass` with runnable status `true`. Mesh-quality preflight is recorded as `mesh_quality_preflight_pass`. The diagnostic r2 variant used 72 target physical elements and 152 target element/IP pairs. Its deck comparison is `scientific_keyword_equivalence_pass`; its static validation is `diagnostic_static_validation_pass`; its runnable flag is `diagnostic_runnable: true`.

A.7 Trace Provenance

The SDV15 diagnostic r2 raw trace is intentionally not stored in Git. It is retained in the HPC evidence area:

```
/home/pr21vyici/projects/adaptive-remeshing/runs/hpc/paper_matched_single_notch_v2_
sdv15_diagnostic_r2/evidence/1375028.mmamaster02/molnar_v2_sdv15_diagnostic_call_trace.
csv
```

The trace size is 258449576 bytes, modified at 2026-07-16T13:24:09+0200, with SHA-256 5b4922884ea6ee7af759c469f6. The completed-increment replay processed 627304 trace rows, retained 209152 U1 stage-101 rows, formed 101840 final increment states, and monitored 152 element/IP sequences.

A.8 Postprocessing Selection Rule and Tolerances

The completed-increment replay keeps the last U1 stage-101 call for every (KSTEP, KINC, physical element, source — storage ID). Retained-frame events are aligned to trace increments by step time/load level, not raw retained frame number.

The replay decrease tolerance is 10^{-8} . The severity-audit materiality tolerance is 10^{-6} . The Fig. 7 digitization uncertainty is recorded as ± 0.00004 mm in displacement and ± 0.015 kN in reaction force. These are evidence definitions, not final supervisor-approved thesis tolerances.

A.9 Committed Evidence Paths

- Baseline run manifest: runs/hpc/paper_matched_single_notch_v2/RUN_MANIFEST.md
- Baseline run summary: runs/hpc/paper_matched_single_notch_v2/RUN_SUMMARY.md
- Technical evidence: runs/hpc/paper_matched_single_notch_v2/evidence/
- Extracted RF-U and contour states: runs/hpc/paper_matched_single_notch_v2/extracted/
- Scientific decision: runs/hpc/paper_matched_single_notch_v2/scientific_review/SCIENTIFIC_DECISION.md
- Fig. 7 comparison metrics: runs/hpc/paper_matched_single_notch_v2/scientific_review/fig7_comparison_metrics.json
- Crack threshold metrics: runs/hpc/paper_matched_single_notch_v2/scientific_review/crack_path_threshold_metrics.csv
- SDV irreversibility metrics: runs/hpc/paper_matched_single_notch_v2/scientific_review/sdv_irreversibility_metrics.json
- Diagnostic r2 summary: runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/RUN_SUMMARY.md
- Completed-increment replay evidence: runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/evidence/1375028.mmamaster02/postprocessing_completed_increment_replay_time_aligned/
- Severity audit: runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/evidence/1375028.mmamaster02/postprocessing_completed_increment_replay_time_aligned/severity_audit/

A.10 Excluded Files

The Abaqus ODB and raw diagnostic trace are intentionally excluded from Git. The candidate-v2 ODB remains on scratch at:

```
/scratch/pr21vyici/adaptive-remeshing/runs/molnar_paper_matched_single_notch_v2_1374864.
mmamaster02/molnar_paper_matched_single_notch_v2.odb
```

The raw diagnostic trace remains in the HPC evidence area and is referenced by hash in this appendix. LaTeX build products such as .aux, .fdb_latexmk, .fls, .out, generated PDFs, and .synctex.gz are local build artifacts.